Vikram Raj Nagoor Kani

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Education

University of Illinois Urbana Champaign — Urbana, IL
Master of Engineering in Autonomy and Robotics — *Aug 2024 - present*

BS Abdur Rahman Crescent Institute of Technology — Chennai, India
Bachelor of Technology in Electronics and Communication Engineering — *Jul 2017 - Jun 2021*

Work Experience

Kohler, *Computer Vision Engineer Intern* — *Jun 2025 - present*
- Developing an **object detection model** to validate packed items during the product inspection process, aiming to **reduce human error** in the production pipeline.

UIUC College of Veterinary Medicine, *AI Engineer Intern* — *Jun 2025 - present*
- Developing a **computer vision pipeline** to evaluate **state-of-the-art** animal **pose estimation models** using our generalized swine dataset, with the goal of identifying the most suitable model for **real-world farm applications**.

UIUC Health and Kinesiology Lab, *Robotics Engineer Intern* — *Jan 2025 - May 2025*
- Integrated **STRETCH AI** with **low-end GPUs**, **optimizing performance** for deployment on STRETCH robot.
- Improved the STRETCH **Robot AI** pipeline for object pick-and-place tasks **enhancing reliability and safety** for elderly assistance and increasing pickup success rate from **20% to 40%**.

Accenture, *Automation Engineer* — *Aug 2021 - Jul 2024*
- Led deployment and testing of **cognitive vision models** on **IoT edge** servers using **Docker containers** to predict, identify and report faulty surveillance cameras at Microsoft Datacenters.
- Developed and led **automation development**, implementing **CI/CD pipelines** for **Azure Cloud resources** and **BareMetal physical servers** procurement, setup and maintenance. **Reducing manual hours spent by 90%**.

Nokia, *Embedded System Engineer Intern* — *Feb 2021 - May 2021*
- Developed an **IoT device** to **detect obstacles** blocking accessibility of fire extinguishers and alert security in **real time** for Nokia's Manufacturing Factory as part of their safety measures.

Projects

LiDAR-Camera Fusion for Robust Navigation
ROS, Python, Open CV, NVIDIA Jetson, F1 Tenth, Unity, Linux
- Introduced a **light weight** lane detection algorithm using **OpenCV**, optimizing **computational speed by 30%** on **Nvidia Jetson** for **real-time waypoint generation** and accepted as **open source contribution** for UIUC ECE 484.
- Developed a **photorealistic digital twin** of the **F1TENTH** car and environment using a Unity-based open-source simulator, enabling testing and validation with **minimal sim-to-real gap** and accelerating development.

6-DOF Robot Arm Pose Prediction Using Deep Learning
PyTorch, Python, OpenCV, ResNet, YOLO
- Developed a **deep learning model** to predict **6 Degrees of Freedom** for **robotic arms** to pick up warehouse parts. Using a combination of RGB and Depth images in a **dual-stream U-Net** architecture with hybrid **feature fusion**.

Persistent Pedestrian Detection Using Sensor Fusion
GEM E4, PyTorch, ROS, Python, Open CV, YOLO, GEMStack
- Developed **persistent pedestrian detection algorithm** utilizing camera and LiDAR data from GEM e4 vehicle optimized for **real-time systems** by implementing techniques like **voxel down-sampling** reducing time complexity.

Publications

Design of Restaurant Service Robot for Contact less and Hygienic Eating Experience — *IRJET 2020*
VIKRAM RAJ.N, *Prejitha.CT, Harshavardhan Vibhandik, et al.*
- Published a paper on theoretical working model of a robot working in a restaurant to prevent spread of COVID-19.

Technical Skills

Programming Languages : Python, C++, Bash, PowerShell, LabView, Matlab, Azure CLI.
Technologies/Frameworks : Pytorch, Open CV, ROS, ROS2, Docker, Roboflow, Anaconda , NVIDIA Jetson, GEM E2, GEM E4, F1Tenth, Azure, Gazebo, Gazebo Ignition, Unity, Git, Linux, YOLO, Detic, SigLIP, Open AI Gym, Deep Learning, Machine Learning, UR3e, CI/CD, SAM, Transformers, ResNet, FANUC.
Coursework : Principles of Safe Autonomy, Deep Learning with Computer Vision, Autonomous Vehicle System Engineering.

Leadership / Extracurricular

Institution of Electronics and Telecommunication Engineers Club, Vice-President
- Proactively organized and led a variety of events for both internal and external college communities. Participated and earned 1st and 2nd place awards in technical coding and quiz competitions.